

SS-RESfM vs. RESfM: Short Summary

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The summary

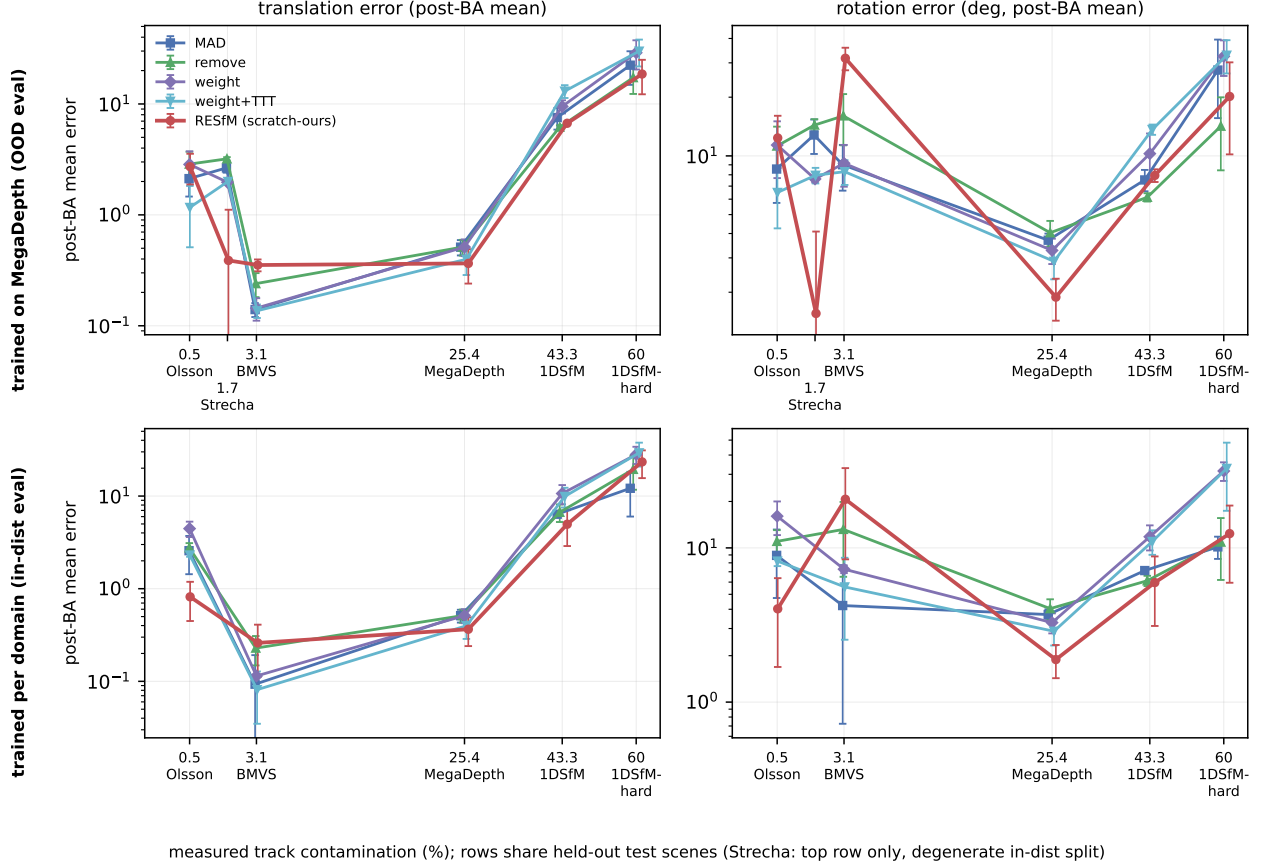
SS-RESfM is RESfM with the supervision removed and nothing else changed — same sets-of-sets architecture, tracks, schedule, and robust bundle adjustment, but the outlier head trains on confident pseudo-labels from the model’s own reprojection-error percentiles instead of COLMAP-derived labels — and the comparison splits along the contamination axis. In its curated comfort zone, supervised RESfM keeps at most a modest edge: in-distribution MegaDepth at 25% contamination, the baseline scores 0.37 ± 0.12 versus 0.40 ± 0.11 for our test-time-fine-tuned arm (statistically overlapping, though that arm carries run-to-run noise) or 0.51 ± 0.08 for the deterministic arms (a $\sim 38\%$ gap); at 43% with best-case labels the bands overlap (5.0 ± 2.1 vs. 6.4 ± 0.2); clean Strecha goes to supervision on every seed (0.39 ± 0.72 vs. $\sim 2\text{--}3.2$); and truly clean *in-distribution* data goes to supervision too (retrained on the Olsson domain at 0.5%: 0.82 ± 0.37 vs. $2.3\text{--}2.6$ for the best label-free arms) — so SS-RESfM does *not* always win: with sound labels and negligible-or-curated contamination, supervision earns its keep. But it matches supervision at moderate out-of-distribution contamination, beats it outright where labels cannot be manufactured (plain removal below *every* baseline seed pair at $\sim 60\%$: 12.4 ± 2.1 vs. 19.2 ± 3.4 , and $2\times$ better even when RESfM is retrained *on* a 60% domain with reference-quality labels — proving the failure is caused by labeling, not domain shift), and wins two of the three clean out-of-distribution splits by $2.5\text{--}2.9\times$ (BlendedMVS and Olsson, where every supervised variant lands at $7.4\text{--}9.1$ versus $2.9\text{--}3.6$ for every label-free arm) with a qualitative difference in dependability: the label-free arms land in a narrow band on every training seed, whereas supervised RESfM collapses on clean OOD in most seeds under every learning rate, budget, and recipe variant tested. The one-line version: removing RESfM’s labels costs little-to-nothing inside its curated comfort zone (and Strecha), and buys robustness — and above all *reliability* — everywhere labels are unsound or unavailable, which is exactly where robust SfM is needed.

What “RESfM collapses” means

“Collapse” is not divergence or a crash — the network trains normally and reconstructs its training distribution well. It is a *silent generalization failure on the easiest possible test*: Olsson, 39 essentially outlier-free scenes (0.5% measured contamination), where a method that merely ignored its outlier head should do fine. Every MegaDepth-trained *supervised* variant we test — the authors’ released checkpoint (9.10), the authors-faithful retrain (8.77), and our controlled from-scratch retrains (7.4 ± 2.6) — lands at a translation error of $7.4\text{--}9.1$, versus $2.9\text{--}3.6$ for every label-free arm and 3.15 for classical GLOMAP. It is seed-dependent but dominant (4 of 5 training seeds collapse; one escapes to 2.32), and it survives every learning rate, epoch budget, and recipe variant we try. Crucially it is *not* a domain-shift story: RESfM retrained in-distribution on the Olsson domain is fine (0.82 ± 0.37 across two seeds) — the supervised outlier head, fit to MegaDepth’s label distribution, actively misclassifies on clean out-of-distribution scenes, while the self-supervised head adapts because its pseudo-labels

are recomputed from the test-time reprojection-error distribution itself.

The picture in one figure



Columns: translation / rotation (post-BA means, log scale). *Top row:* all methods trained on MegaDepth, evaluated out-of-distribution on each domain’s held-out test scenes. *Bottom row:* trained on each domain itself (supervised with best-case GT-derived labels), evaluated in-distribution on the same scenes. The red supervised line is RESfM-from-scratch (our controlled anchor) in the top row and the in-domain retrain in the bottom row (MegaDepth anchor point shared); label-free arms carry five-seed bands. In-domain retraining rescues supervision at 0.5% and helps at 43%, does nothing at 3.1% (the dead zone — rotation-first in both rows), and worsens at 60%, while the label-free arms barely move between rows: they have no labels to gain from. The top row also shows Strecha (1.7%, full four-scene OOD eval; no bottom-row counterpart — its same-protocol in-distribution split leaves a single uninformative test scene): supervision’s clean-OOD win there is visible in both columns. Analogous figures exist with the authors-faithful retrain and the released checkpoint as the supervised line; the verdicts are unchanged. (Test-scene restriction caveats: the 8 Olsson held-out scenes exclude most of the clean-OOD collapse scenes, and the 43% point is a three-scene mean including the Ellis Island failure scene.)

Why five seeds (seed-additivity fails)

If the training seed shifted all models equally, relative differences would be seed-invariant and one seed would suffice — the implicit assumption of prior work in this benchmark family, which fixes a single seed. Tested directly on the saved per-scene results (98 scene-cells, five seeds), the assumption fails: the sign of the supervised-vs-MAD difference flips across seeds in 67% of scenes, the across-seed spread of the difference matches or exceeds the margin itself, and the *dataset-level* verdict flips across seeds on four of six datasets (MegaDepth, 1DSfM, 1DSfM-hard, and Olsson — where the one escape seed briefly hands supervision the win); only Strecha and BlendedMVS are seed-invariant, and those are exactly the two cells stated as every-seed claims. A single-seed study can therefore reverse the headline conclusion on most datasets — which is why every comparison in this project carries five-training-seed bands, and the strongest claims are verified over all seed pairings. (Evaluation seeds, by contrast, are a shared negligible offset, ± 0.008 — replicating over them is unnecessary, and we don’t.)

The four RESfM reference points

All four supervised reference points appear in the paper’s main table; translation error (mean), lower is better. The controlled comparison arm is RESfM-scratch: identical tracks, schedule, and BA to our label-free arms, with checkpoint selection by our label-free reprojection-error criterion (which *favors* the baseline relative to the authors’ own Accuracy-based selection — MegaDepth 0.37 vs. 0.50).

RESfM variant	MegaDepth (25%)	1DSfM (43%)	1DSfM-hard (60%)	Strecha (1.7%)	BMVS (3.1%)	Olsson (0.5%)
Paper (reported)	0.169	–	–	–	–	–
Released (authors’ weights)	0.203 $\pm$ 0.008	10.71	15.29	0.20	0.33	9.10
From scratch, faithful (authors’ ckpt sel.)	0.496	9.75	17.58	0.14	0.37	8.77
From scratch, our reproj-error ckpt sel.	0.37 $\pm$ 0.12	10.5 $\pm$ 1.5	19.2 $\pm$ 3.4	0.39 $\pm$ 0.72	0.35 $\pm$ 0.04	7.4 $\pm$ 2.6
SS-RESfM best label-free arm	0.40 $\pm$ 0.11	8.9 $\pm$ 1.2	12.4 $\pm$ 2.1	1.97 $\pm$ 0.09	0.14 $\pm$ 0.02	2.9 $\pm$ 0.3

Reading the four rows: the *paper* number (0.169) is not reproducible even from the authors’ own released weights in our harness (0.209, confirmed under their pinned software stack); the *released* checkpoint is the strongest in-distribution but collapses on Olsson like every other supervised variant (9.10) — so the collapse is not an artifact of our retraining; the *faithful* from-scratch retrain (authors’ own checkpoint-selection metric) is the weakest in distribution (0.496), showing our selection deviation helps the baseline, not us; and *from-scratch with our selection* is the controlled anchor used for all seed-band claims. Against whichever row one prefers, the label-free story is unchanged: supervision holds MegaDepth and Strecha; label-free wins the high-contamination regime, BlendedMVS, and Olsson, and is the only recipe that never collapses.